A Wideband 1-Bit Reconfigurable Reflectarray Antenna at *Ku*-Band

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***Abstract*—In this letter, a wideband high-efficiency 1-bit 16 × 16 reconfigurable reflectarray antenna (RRA) is proposed. First, a wideband and low loss 1-bit RRA element is designed. It consists of double-layer patches printed on two dielectric substrates, respectively, forming a multiresonant behavior to expand the bandwidth. Due to the relatively thin thickness of the double-layer dielectric substrate (0.05 λ_0), the element has low reflection loss. Therefore, the high efficiency and wide bandwidth performances can be obtained simultaneously. The phase difference between the two states of the RRA element is stable within 180 ± 20° in the frequency range of 12.9–16.5 GHz. Based on the broadband RRA element, a 16 × 16 RRA was fabricated and measured. Tested results show that the proposed RRA achieves 22.5% of 1 dB gain bandwidth and 25% of maximum aperture efficiency.**

***Index Terms*—Beam scanning, high efficiency, p-i-n diode, reconfigurable reflectarray, wideband.**

I. Introduction

In THE fields of satellite communications, such as space explorations and mobile communication, in order to adapt to the long-distance communication and relatively complicated space environment, it is essential to design high-gain antennas. As a new generation of high-gain antenna, the reconfigurable reflectarray antenna (RRA) [1] combines the beam-scanning characteristic of phased array antenna based on the traditional reflectarray antenna [2], [3]. Besides, it also has a lot of merits of low profile, light weight, low cost, and easy fabrication. The beam agile capability of RRA can meet the requirements of automatic scanning in satellite communications. As a high-gain directional antenna, the RRA has great potential in the applications of radar and communication systems.

RRA has many advantages, and it has been well verified that the reflectarray antenna can realize the reconfigurable function through lumped elements [4]–[9]. However, they all have the disadvantage of narrow bandwidth [10]. In order to increase the information transmission rate of satellite communications, it is necessary to enhance the operating bandwidth of the antenna. For small and medium sized RRAs, the bandwidth of the reflectarray element is the main factor limiting the gain bandwidth of RRA. In terms of the element structure, the common ways to increase bandwidth include a single-layer multiresonance structure [11] and a multilayer stacked structure [12]. However, for the RRA, the single-layer multiresonance structure is not suitable for the connection between the phase-shifting structure and lumped components, which increases the difficulty of design and fabrication. Although the bandwidth can be broadened by increasing the thickness of the dielectric substrate, the thick dielectric will strongly excite the surface waves and some high-order modes, which will increase the element loss and reduce the radiation efficiency. For satellite digital broadcasting, the *Ku*-band is more conducive to the transmission of digital signals compared with the *C*-band due to its better stability. However, the higher frequencies of *Ku*-band bring in more losses than *C*-band, and the loss of lumped element is increased at the same time. In addition, the errors caused by phase quantization will further reduce the aperture efficiency of the RRA. Therefore, it has important significance to investigate broadband, high-efficiency *Ku*-band RRA.

Up to now, there have been relatively few studies on *Ku*-band broadband high-efficiency RRA. Two resonant structures are employed to broaden the bandwidth of the varactor diode-tuned element in [13]. By using a square slotted patch loaded with a p-i-n diode and optimizing the phase-shift distribution at the center frequency, the RRA achieves wideband characteristics [14]. Although the gain bandwidth of these structures in [13] and [14] has been improved to some extent, their aperture efficiencies are merely 7.5% and 15.2%, respectively. In [15], the aperture efficiency of the antenna is improved by optimizing the feed position and the phase distribution. However, the gain bandwidth of the antenna is still limited. In [16], the aperture efficiency has been improved to 25% by placing p-i-n diode on the back of the RRA, whereas the 1 dB gain bandwidth of the RRA is only 3%.

In this letter, a multilayer stacked structure is used to expand bandwidth. Meanwhile, the p-i-n diode is placed below the ground plane [14], which not only facilitates the connection between the two-layer patches and the p-i-n diode but also improves the aperture efficiency of the RRA. In addition, two thin dielectric substrates are used to reduce the dielectric loss without affecting the working bandwidth. Based on the RRA element, a 1-bit RRA operating in *Ku*-band is proposed. Measured results show that the 1 dB gain bandwidth of the RRA is 22.5%, and the peak aperture efficiency is 25%. The wide-angle beam-scanning performance reaches ±60°. It can be concluded from the measured results that the proposed method effectively improves the gain bandwidth and aperture efficiency for the RRA.

Manuscript received November 20, 2021; accepted December 20, 2021. Date of publication December 28, 2021; date of current version March 3, 2022. This work was supported by the National Key Research and Development Program of China under Grant 2020YFA0713900. *(Corresponding author: Gang Zhao.)*

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Digital Object Identifier 10.1109/LAWP.2021.3138438

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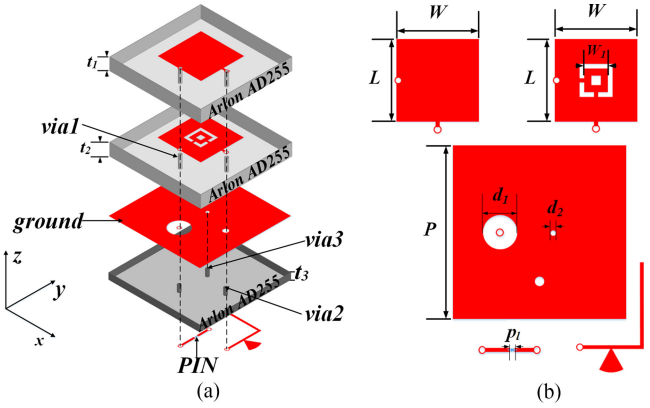


Fig. 1. Topology of the proposed 1-bit RRA element. (a) Perspective view. (b) Top view.

II. Wideband 1-Bit RRA Element Design

In this letter, the multilayer stacked structure is adopted to design the RRA element. Two layers of patches form double resonance behavior, which obtains a smooth and good linear reflection phase curve. The element structure is shown in Fig. 1, which includes three stacked dielectric substrates and four metal layers. Arlon AD255 ($\varepsilon_r = 2.55$ and $\tan\delta = 0.0014$) is used for all three dielectric layers. A rectangular patch and a slotted square patch are printed on the top surface of the upper and middle dielectric layers, respectively. The ground plane is placed between the middle substrate and the bottom substrate, which is conducive to grounding the dc circuits. The lower surface of the bottom dielectric substrate contains the dc bias line and two microstrip lines for connecting p-i-n diode. The two radiating patches are connected through the metalized via holes, one microstrip line is connected to the end of the p-i-n diode and the other one is connected to the other end of the p-i-n diode, which is finally connected to the ground plane. The welding position of the p-i-n diode is shown in Fig. 1(b). The resonant characteristics of the element change correspondingly when the p-i-n diode is ON or OFF under the control of the dc bias voltage, which eventually guarantees the proposed element has a phase difference of 180° between the two states.

The MACOM MADP-000907-14020W p-i-n diode is chosen in this design, which can operate in the desired frequency with low insertion loss. In the element simulation, the p-i-n diode is modeled as the *RLC* boundary condition in the High Frequency Structure Simulator (HFSS) software. For the ON state, the p-i-n diode is modeled as a lumped resistor with a typical value of 5.2 Ω, while the p-i-n diode is modeled as a lumped capacitance with a typical value of 25 fF on the OFF state. The bias point should be designed at the equivalent zero-electric field position of the element to reduce the influence of the dc signal on the reflection field. To isolate the radio frequency signal from the dc signal, an open-ended radial stub and a quarter-wavelength bias line with high resistance are connected to the dc line.

The proposed RRA element is simulated by ANSYS HFSS software with periodic boundary conditions. Some key geometric parameters are: $P = 9.5$ mm, $t1 = t2 = 1.016$ mm, $t3 = 0.508$ mm, $L = W = 4.5$ mm, and $W1 = 1.26$ mm. Under the condition of vertical incidence, the reflection phase and amplitude results of the RRA element are simulated in the ON and OFF states, as shown in Fig. 2(a). It can be obtained from the figure that a phase difference of 180° ± 20° is achieved

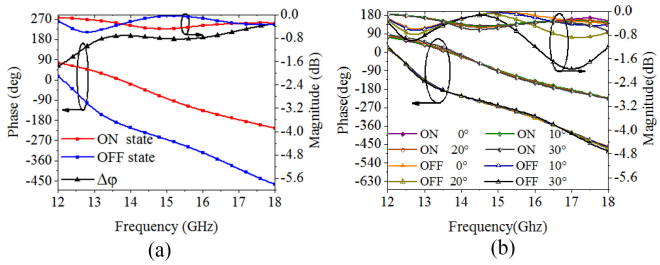


Fig. 2. Simulated reflection coefficients of the proposed RRA element. (a) Vertical incidence. (b) Oblique incidence.

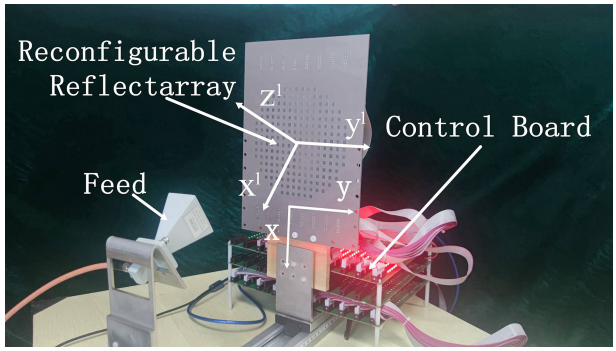


Fig. 3. Photograph of the proposed RRA.

from 12.9 to 16.5 GHz, and the relative bandwidth is 23.8%. The proposed RRA element has good broadband performance. In the operating frequency band, the reflection loss of the element is less than 0.6 dB, which indicates that the double-layer dielectric plate does not increase the loss. It is beneficial to improve the aperture efficiency of the RRA. At the same time, the element characteristics for oblique incidences are given in Fig. 2(b). It can be seen from the figure that the phase curves of the RRA element proposed in this letter with different incidence angles are almost consistent, and the amplitudes change a little within the working bandwidth.

III. 16 × 16 RRA Prototype Design and Measurement

A. 16 × 16 RRA Design

Based on the proposed broadband 1-bit RRA element, a 16 × 16 RRA is designed and fabricated, as shown in Figs. 3 and 4. The RRA array is arranged in a circular pattern, which consists of 208 reflectarray elements with an effective aperture diameter of 152 mm, corresponding to 7.6 λ_0 at 15 GHz. A total of 16 sockets are arranged on both sides of the array to connect to the main control board. Every socket contains 16 dc input channels corresponding to the 16 elements in two adjacent columns on one side of the RRA array. The size of the printed circuit board is 271 × 192 mm^2. Each element of the reflectarray contains one p-i-n diode, and the total of 208 p-i-n diodes is soldered on the RRA. Since the RRA is *y*-polarization, in order to reduce the influence of the dc biasing lines on the radiation performance of the antenna, the dc biasing lines are placed in the *x*-axis direction.

B. Control Board

The main control board is an important part of the RRA design, and its prototype is shown in Fig. 3. The main control

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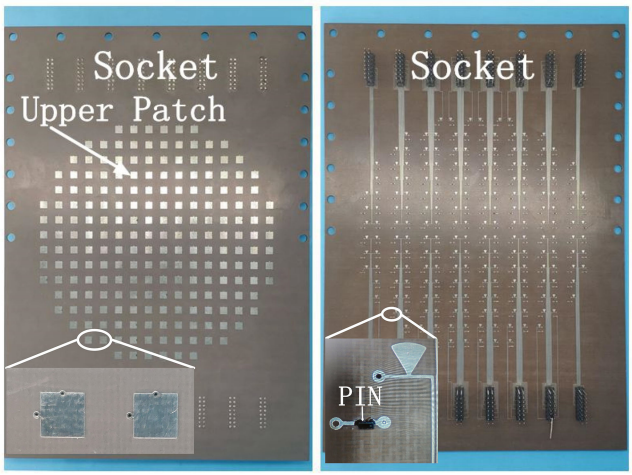


Fig. 4. Layout of biasing lines on the bottom layer of the proposed RRA.

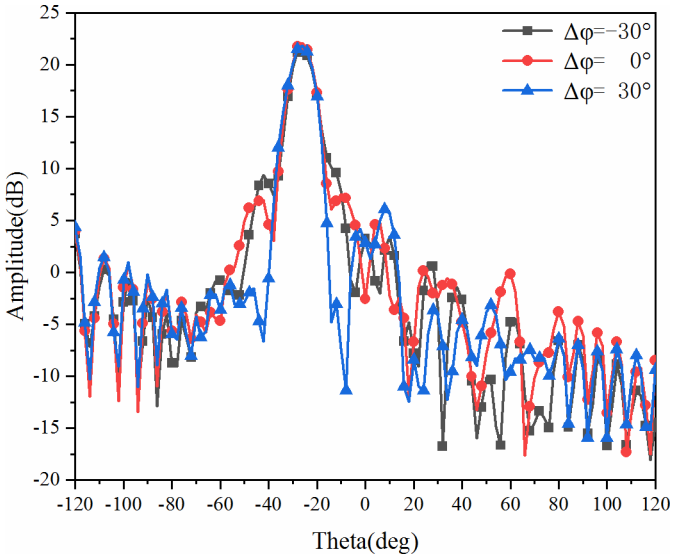


Fig. 5. Simulated radiation patterns with different reference phases.

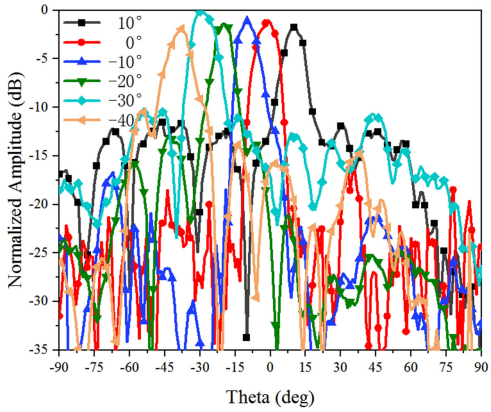


Fig. 6. Measured radiation patterns for the main beam scanned from −40° to 10° with a step of 10° in the *xoz* plane at 15 GHz.

board is designed based on microcontroller unit (MCU) components, which outputs 256 signals in parallel. It can independently control the voltage of each dc input and then control the ON or OFF of each p-i-n diode. At the same time, an LED display circuit is added to each output terminal, which can display the phase distribution of the array. It can check the correctness of the output of the control circuit.

C. Feed

In this letter, a standard pyramid horn at the *Ku*-band is chosen as feed, which can emit the *y*-polarized spherical wave. Its equivalent phase center is 14 mm behind the aperture plane of the horn, and the 10 dB beamwidth is 52°. Therefore, the focal diameter ratio is 1.03, and the distance from the equivalent phase center of the horn to the RRA is 156.6 mm. In order to reduce the blocking effect, the offset feeding method is adopted. The horn is located in the *xoz* plane, and the offset angle is 26°.

D. Optimization of Compensation Phase Distribution

For the reflectarray, the spherical wave emitted by the feed is converted to the plane wave in the (θ_0, φ_0) direction, and the phase shift of the *mn*th element is calculated as follows:

$$\varphi_{r_{mn}} = k\left[d_{mn} - \sin\theta_0\left(x_{mn}\cos\varphi_0 + y_{mn}\sin\varphi_0\right)\right] + \Delta\varphi \tag{1}$$

where $\Delta\varphi$ is a constant reference phase [17], k represents the propagation constant, and d_{mn} is the distance between the *mn*th element (x_{mn}, y_{mn}, z_{mn}) and the feed phase center (x_f, y_f, z_f).

According to (1), the continuous phase-shift compensation of the reflectarray is within the range of 0°–360°. Since there are only two states for each element, the 1-bit discrete phases are required. In this design, the compensation phases of the two states at 15 GHz are designed as 90° and 270°, and the standard 180° phase shift is selected as the quantization standard. The quantization method is shown as follows:

$$\varphi_{r_{mn}} = \begin{cases} 90^\circ, 0 \le \varphi_{r_{mn}} < 180^\circ \\ 270^\circ, 180^\circ \le \varphi_{r_{mn}} < 360^\circ. \end{cases} \tag{2}$$

In fact, the use of discrete phases leads to a degradation of the radiation performance of the reflectarray, such as gain decrease, high sidelobe level, and main beam direction error. Although these adverse effects can be reduced by increasing the number of RRA elements, they also can be improved by optimizing the reference phase $\Delta\varphi$ when the array size is determined. In this design, the radiation patterns corresponding to different reference phases are simulated when the direction of the main beam is −26° in the *xoz* plane, as shown in Fig. 5. It can be seen that the different sidelobe levels will be generated when the antenna adopts different reference phases, and the best reference phase of the main beam at −26° is 30°, and the sidelobe levels are less than −15 dB. Therefore, in order to improve the radiation performance, it is necessary to optimize the reference phase for each beam-scanning direction θ_0.

E. Beam-Scanning Design and Measurement

Since the p-i-n diode on each RRA element can be independently controlled through the main control board, the phase of each element can be adjusted flexibly, and the beam-scanning performance can be achieved. As the coordinate system of $x^1y^1z^1$ is shown in Fig. 3, the direction of the maximum gain of the antenna in this design is z^1-axis, which is oriented at 30° to the *z*-axis. Figs. 6 and 7 give the measured beam-scanning results in the *xoz* and y^1oz^1 planes at 15 GHz, respectively. Due to

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TABLE I
COMPARISON OF THE PROPOSED RRA WITH SOME EXISTING WORKS

Ref.	[4]	[6]	[9]	[13]	[14]	[15]	[16]	**This work**
Freq. (GHz)	60.25	11.1/14.3	12	5.4	5	12.5	11.5	**15**
Size	112λ×112λ	20λ×20λ	NA	NA	6.6λ×6.6λ	5λ×5λ	5.1λ×5.1λ	**7.6λ×7.6λ**
Lumped Component Type	PIN	PIN	MEMS	Varactor	PIN	PIN	PIN	**PIN**
Element BW (%)	0.75	3.2/4.8	15	2.4	12	4	12.4	**23.8**
1dB Gain BW (%)	NA	4.5/6.2	NA	NA	8.4	6	NA	**22.5**
Gain (dBi)	42	29.3/30.8	NA	NA	19.22	17.5	19.2	**21.6**
Aperture Efficiency (%)	9.5	21.6/18.1	NA	NA	15.26	17.9	25	**25**

Note that NA refers to not available.

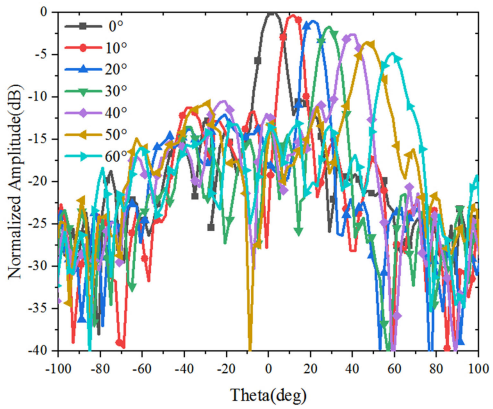


Fig. 7. Measured radiation patterns for the main beam scanned from 0° to 60° with a step of 10° in the y^1oz^1 plane at 15 GHz.

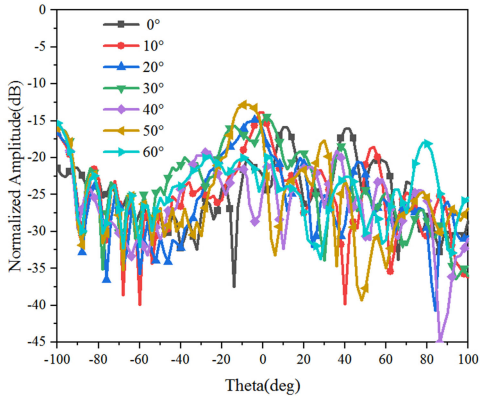


Fig. 8. Measured cross polarization in the y^1oz^1 plane at 15 GHz.

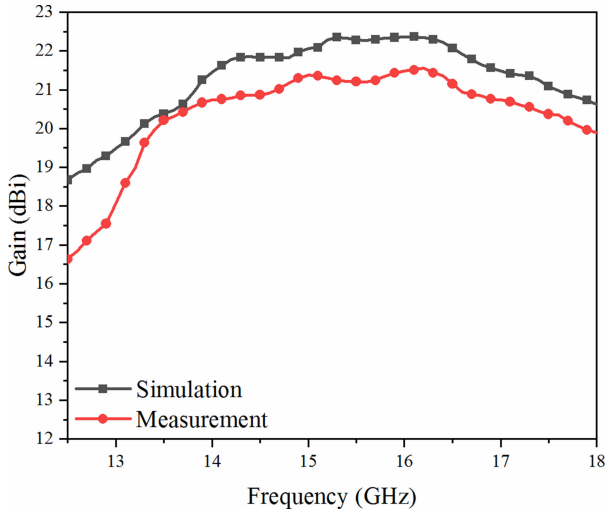


Fig. 9. Simulated and measured gain of the proposed RRA.

the blocking effect of the offset feed, it is found that the RRA can achieve the best beam-scanning function in the range of −40°–10° in the *xoz* plane. Since there is no blockage caused by the feed in the y^1oz^1 plane, the sidelobe levels are much better than those in the *xoz* plane, and scan angles of ±60° can be achieved. As shown in Fig. 7, the gain difference of the main beams between 0° and 60° directions is less than 5 dB, and the sidelobe level of the case that the main beam point at 0° is below −10 dB. With the increase of beam-scanning angle, the sidelobe level of RRA increases slightly, and the gain decreases. The cross polarizations of this design are shown in Fig. 8, and their levels are less than −13 dB. It can be obtained that the RRA has good two-dimensional beam-scanning performance.

In addition, the gain versus frequency with +30° beam steering in the *xoz* plane is studied, and the corresponding simulated and tested results are shown in Fig. 9. The measured maximum gain of the RRA is 21.6 dB, corresponding to the aperture efficiency of 25%. The 1 dB gain bandwidth of the RRA is 22.5% (13.7–16.3 GHz). Compared with the RRAs in the literature in Table I, it can be seen that the proposed RRA element has a wider 1 dB gain bandwidth and higher aperture efficiency.

IV. CONCLUSION

In this letter, a wideband high-efficiency 1-bit RRA with 16 × 16 elements operating in *Ku*-band is proposed. In this design, the RRA element expands the bandwidth by stacking two layers of radiating patches to form two resonant frequencies close to each other. Meanwhile, the reconfigurable element in this design does not increase the reflection loss by using the stacking structure, which is conducive to improving the aperture efficiency of RRA. The measured results show that the proposed RRA has a 1 dB gain bandwidth of 22.5% and an aperture efficiency of 25%.

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REFERENCES

[1] P. Nayeri, F. Yang, and A. Z. Elsherbeni, "Beam-scanning reflectarray antennas: A technical overview and state of the art," *IEEE Antennas Propag. Mag.*, vol. 57, no. 4, pp. 32–47, Aug. 2015.

[2] D. Berry, R. Malech, and W. Kennedy, "The reflectarray antenna," *IEEE Trans. Antennas Propag.*, vol. AP-11, no. 6, pp. 645–651, Nov. 1963.

[3] P. Nayeri, F. Yang, and A. Z. Elsherbeni, *Reflectarray Antennas: Theory, Designs, and Applications*. New York, NY, USA: Wiley, 2018.

[4] H. Kamoda, T. Iwasaki, J. Tsumochi, T. Kuki, and O. Hashimoto, "60-GHz electronically reconfigurable large reflectarray using single-bit phase shifters," *IEEE Trans. Antennas Propag.*, vol. 59, no. 7, pp. 2524–2531, Jul. 2011.

[5] E. Carrasco, M. Barba, and J. A. Encinar, "X-band reflectarray antenna with switching-beam using PIN diodes and gathered elements," *IEEE Trans. Antennas Propag.*, vol. 60, no. 12, pp. 5700–5708, Dec. 2012.

[6] H. Yang *et al.*, "A 1600-element dual-frequency electronically reconfigurable reflectarray at X/Ku-band," *IEEE Trans. Antennas Propag.*, vol. 65, no. 6, pp. 3024–3032, Jun. 2017.

[7] L. Boccia, G. Amendola, and G. Di Massa, "Performance improvement for a varactor-loaded reflectarray element," *IEEE Trans. Antennas Propag.*, vol. 58, no. 2, pp. 585–589, Feb. 2010.

[8] H. Rajagopalan, Y. Rahmat-Samii, and W. A. Imbriale, "RF MEMS actuated reconfigurable reflectarray patch-slot element," *IEEE Trans. Antennas Propag.*, vol. 56, no. 12, pp. 3689–3699, Dec. 2008.

[9] J. Perruisseau-Carrier and A. K. Skrivervik, "Monolithic MEMS-based reflectarray cell digitally reconfigurable over a 360° phase range," *IEEE Antennas Wireless Propag. Lett.*, vol. 7, pp. 138–141, 2008.

[10] D. M. Pozar, "Bandwidth of reflectarrays," *Electron. Lett.*, vol. 39, no. 21, pp. 1490–1491, 2003.

[11] J. H. Yoon, Y. J. Yoon, W.-S. Lee, and J.-H. So, "Broadband microstrip reflectarray with five parallel dipole elements," *IEEE Antennas Wireless Propag. Lett.*, vol. 14, pp. 1109–1112, 2015.

[12] J. A. Encinar and J. A. Zornoza, "Broadband design of three-layer printed reflectarrays," *IEEE Trans. Antennas Propag.*, vol. 51, no. 7, pp. 1662–1664, Jul. 2003.

[13] C. Liu and S. V. Hum, "An electronically tunable single-layer reflectarray antenna element with improved bandwidth," *IEEE Antennas Wireless Propag. Lett.*, vol. 9, pp. 1241–1244, 2010.

[14] J. Han, L. Li, G. Liu, Z. Wu, and Y. Shi, "A wideband 1 bit 12 $\times$ 12 reconfigurable beam-scanning reflectarray: Design, fabrication, and measurement," *IEEE Antennas Wireless Propag. Lett.*, vol. 18, no. 6, pp. 1268–1272, Jun. 2019.

[15] H. Yang *et al.*, "A 1-bit 10 $\times$ 10 reconfigurable reflectarray antenna: Design, optimization, and experiment," *IEEE Trans. Antennas Propag.*, vol. 64, no. 6, pp. 2246–2254, Jun. 2016.

[16] H. Zhang, X. Chen, Z. Wang, Y. Ge, and J. Pu, "A 1-bit electronically reconfigurable reflectarray antenna in X band," *IEEE Access*, vol. 7, pp. 66567–66575, 2019.

[17] A. H. Abdelrahman, P. Nayeri, A. Z. Elsherbeni, and F. Yang, "Bandwidth improvement methods of transmitarray antennas," *IEEE Trans. Antennas Propag.*, vol. 63, no. 7, pp. 2946–2954, Jul. 2015.

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